Program : **Diploma in Polymer Technology**	
Course Code : **6091C**	Course Title: **Polymer Physics**
Semester : **6**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester: **60**

Course Objectives:

- To understand about the relation between structural features of polymers and their properties.
- To impart knowledge on thermal transitions associated with polymers
- To understand rheological properties of polymers

Course Prerequisites:

Topic	Course code	Course name	Semester
Polymer science		Polymer science	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize various structural features associated with polymeric molecules	13	Understanding
CO2	Correlate structure and mechanical properties of polymers	15	Understanding
CO3	Explain the basic concepts of rheology of polymers	15	Understanding
CO4	Explain various thermal transitions and their determination.	15	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Summarize various structural features associated with polymeric molecules**		
M1.01	Outline the concepts details associated with microstructure of polymers	3	Remembering
M1.02	Comprehend the effect of various factors related to polymerization on the molecular structure of polymer	3	Understanding
M1.03	Summarize fundamental difference between crystalline state and amorphous state in polymers.	4	Understanding
M1.04	Enlist various factors affecting grass transition and crystallization.	3	Understanding
Contents: Chemical bonding and polymer structure-primary structure, secondary structure, configuration, diene polymerization, tacticity, conformation, molecular weight, secondary bonding forces, crystalline and amorphous structure of polymers, crystallization tendency, factors affecting glass transition and crystallization			
CO2	**Correlate structure and mechanical properties of polymers**		
M2.01	Summarize the terminologies associated with mechanical properties of polymers.	3	Remembering
M2.02	Describe mechanical tests performed in polymers	4	Remembering
M2.03	Explain the factors affecting mechanical properties of polymers	4	Understanding

M2.04	Outline methods for improving mechanical properties of polymers	4	Understanding
	Series Test – I	1	

Contents:

Mechanical properties of polymers, Mechanical tests, stress-strain behavior, creep, stress relaxation, dynamic mechanical analysis, impact, factors affecting mechanical properties, effect of molecular weight, cross-linking, crystallinity, copolymerization, plasticizers, polarity, steric factors, temperature, strain rate –common methods for modifying mechanical properties of polymers.

CO3	**Explain the basic concepts of rheology of polymers**		
M3.01	Compare the response of a solid and liquid to an applied force.	4	Understanding
M3.02	Explain various types of fluids	3	Remembering
M3.03	Illustrate Maxwell and Berger model	4	Understanding
M3.04	Discuss the response of polymer melt towards dynamic loads.	4	Understanding

Contents:

Definition of rheology, response of solids and liquids to mechanical forces, different types of fluids, polymer viscoelasticity, mechanical models for linear viscoelastic response, Maxwell and Voight models, response of polymer melt to dynamic loading, effect of shear stress, effect of shear rate, effect of temperature.

CO4	**Explain various thermal transitions and their determination**		
M4.01	Describe different thermal transition associated with polymers	3	Understanding
M4.02	Explain working principle and the determination of thermal transitions using DSC.	4	Understanding
M4.03	Explain working principle and the determination of thermal transitions using DMA.	4	Understanding
M4.04	Explain working principle and the determination of thermal transitions using TMA.	4	Understanding
	Series Test – II	1	

Contents:

Thermal transitions associated with polymers, glass transition, crystallization, melting, Volume changes associated with thermal transitions, specific heat changes associated with thermal transitions, mechanical property changes associated with polymers, Working principle and the determination of thermal transitions using DSC, DMA and TMA

Text / Reference:

T/R	Book Title/Author
T1	Polymer science, V R Gowarikar
R1	Brown R.P.-Physical Testing of Rubbers, Chapman & Hall
R2	Brown R.P-Handbook of plastics Test Methods, Harlow, Longman Scientific and Technical, NY, Wiley 1988
R3	Cowie, J.M.G,Polymers: Chemistry and Physics of Modern material, 2nd Edn., Chapman & Hall, New York
R4	Brydson. J. A - Flow properties of polymer melts - Godwin, Plastic & Rubber Institute.

Online Resources:

Sl.No.	Website Link
1	https://www.tainstruments.com/brochure-library/#1613409887356-b4713171-c466
2	https://en.wikipedia.org/wiki/Polymer_physics
3	https://www.sciencedirect.com/topics/materials-science/polymer-physics